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Investigating Walking Stability of Exoskeleton-Assisted Load Carrying Tasks: A Pilot Comparison of Three Commercial Devices

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Abstract

Maintaining walking stability while carrying loads is important for reducing fall risk and musculoskeletal strain. Although exoskeletons have been increasingly used in industrial settings to reduce work-related musculoskeletal injuries, their effects on whole-body dynamic stability during realistic manual material handling workflows remain insufficiently understood. This study investigated the influence of three exoskeletons on dynamic stability during a simulated load-carrying task. Nine participants performed a simulated manual material handling task involving repeated lifting, carrying a load n to a nearby shelf, and returning to the starting location. Each participant completed this task under four different, order-randomized conditions: No Exoskeleton (NE), and wearing Ekso Bionics Ekso Vest (EV), Levitate AIRFRAME (AF), and suitX ShoulderX (SX). Motion data collected during load-carrying walking were analyzed. Center-of-mass (COM) acceleration in the mediolateral (ML) and anteroposterior (AP) directions was evaluated, and sample entropy (SampEn) was computed to quantify the temporal irregularity of COM acceleration as an indicator of dynamic balance control. Statistical analysis showed no significant differences in SampEn across exoskeleton conditions in either the ML or AP direction at the 0.05 level. However, descriptive trends in the AP direction suggested a marginally significant lower SampEn under the EV condition relative to NE, indicating a possible reduction in the irregularity of COM acceleration. These findings suggest that while no statistically significant exoskeleton effect was detected in this pilot sample, temporal complexity analysis of COM motion may still be useful for characterizing exoskeleton-related changes in locomotor control during load-carrying tasks.

Keywords

occupational exoskeleton, manual material handling, gait stability, center of mass, sample entropy

1. Introduction

Manual material handling (MMH) is ubiquitous in occupational settings and is closely linked to the large global burden of low back pain [1]. To mitigate lumbar loading during work-relevant tasks such as lifting and forward bending, industrial back-support exoskeletons have been increasingly developed and evaluated, with evidence suggesting they can reduce physical workload indicators such as trunk muscle activity in controlled lifting paradigms [2-6]. As exoskeleton deployment expands, there is also a growing need to evaluate their effects under task sequences that better reflect real workflows, rather than isolated lifting alone [7].

Beyond reducing local mechanical demand, exoskeletons may alter whole-body control when workers transition from lifting to walking, leading to falling risks that are insufficiently studied. Because walking stability reflects the organization of motor control over time, condition effects may manifest in the temporal structure of movement signals rather than only in variability magnitude. Therefore, entropy-based analyses, such as approximate entropy, have been broadly applied and synthesized in methodological reviews for gait research, supporting their use as complementary descriptors of locomotor control [8], while also highlighting that entropy outcomes can vary with analysis choices in short or segmented datasets [9]. Sample entropy (SampEn) is a widely used nonlinear metric of time-series regularity that was introduced to improve consistency relative to approximate entropy for physiological data [10]. In locomotion, the COM acceleration reflects the output of neuromuscular control and balance regulation, its moment-to-moment irregularity captures how variable and adaptable the control strategy is over time. SampEn is therefore suitable for

characterizing changes in the temporal structure regularity and complexity, beyond what can be inferred from conventional linear variability measures.

Despite growing evidence that occupational exoskeletons can reduce local biomechanical demands during lifting-related tasks, less is known about how different commercial devices influence whole-body dynamic stability during realistic lift-carry-walk workflows. In addition, most prior evaluations have emphasized local or quasi-static outcomes rather than the temporal organization of gait-related COM dynamics. To address this gap, this pilot study compared three commercial upper-body exoskeletons and a no-exoskeleton condition during a simulated repeated MMH workflow. SampEn of whole-body COM acceleration in the AP and ML directions during walking was used to evaluate whether exoskeleton configuration altered the temporal regularity of locomotor control.

2. Method

2.1. Participants

Nine adults participated in this study (age: 32±3.6 years; height: 177.7±8.6 cm; body mass: 79.91±16.2 kg). All participants were able to walk independently without assistive devices. Participants reported no history of low-back injury or chronic back pain, and no known musculoskeletal, neurological, or vestibular disorders that could affect balance or gait. Additional exclusion criteria included any lower-limb injury within the past few months, prior surgery affecting normal gait, or any condition that would limit safe participation in lifting and walking tasks. All participants provided written informed consent prior to participation, and the experimental procedures were approved by the New Jersey Institute of Technology Institutional Review Board (IRB protocol #2403041907).

2.2. Experimental Protocol

Each participant completed a simulated manual material handling (MMH) task that combined repeated lifting and short bouts of walking while carrying a box-shaped load. At the start of each cycle, the participant lifted a rigid box from a fixed location on the floor using a standardized technique. The box mass was 30 lbs and its dimensions were 14 inch (L) × 10 inch (W) × 6 inch (H). After lifting, the participant turned toward a shelving unit located in front of them and walked forward at a self-selected comfortable speed while carrying the load. Upon reaching the force platform positioned in front of the shelf, the participant placed both feet on the platform, came to a stable stand, and then placed the box onto the shelf. The participant returned to an upright standing posture and paused for approximately 1–2 s, then retrieved the box from the shelf, turned back toward the original starting location, walked back at a self-selected comfortable speed, and finally placed the box back on the floor at the original location, completing one cycle.

Each trial consisted of five consecutive cycles of this lift–walk–place sequence. Participants performed the protocol under four conditions: (1) without an exoskeleton (NE) and (2–4) while wearing three different exoskeleton configurations (EV, AF, SX). For each condition, two trials were completed, resulting in a total of eight trials per participant. Prior to data collection, participants were fitted with reflective markers according to the study's full-body marker set, and all trials were performed using the same task instructions and setup shown in Figure 1.

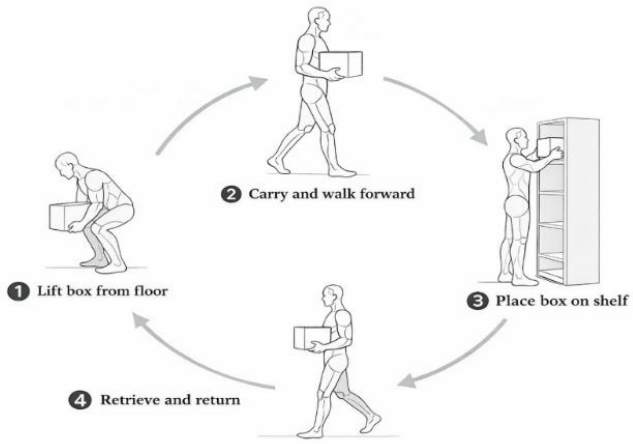


Figure 1: Schematic illustration of the experimental protocol for one cycle of the manual material handling task, consisting of lifting a box from the floor, walking forward while carrying the load, placing the box onto a shelf, and returning the box to the original location.

2.3. Instrumentation and Data Acquisition

Whole-body kinematics were recorded using an OptiTrack motion-capture system (NaturalPoint, Inc., Corvallis, OR, USA) at 100 Hz. Participants were instrumented with a full-body reflective marker set, and marker trajectories were labeled and processed in Motive 3. Whole-body center-of-mass (COM) kinematics were estimated in OpenSim v4.4 using a subject-specific scaled musculoskeletal model. Marker trajectories exported from Motive were imported into OpenSim to scale the generic model to each participant and to perform inverse kinematics analysis. The whole-body COM trajectory was then computed from the modeled segment kinematics and exported for post-processing.

2.4. Signal Processing

COM position time series were exported from the motion-capture pipeline at 100 Hz and processed in MATLAB. The mediolateral (ML) and anteroposterior (AP) COM positions were extracted. COM velocities and accelerations were computed using a four-point central-difference scheme implemented as Equation 1 and 2 shown below. Boundary samples were handled by padding the first two and last two frames with the nearest valid values.

$$v(k) = \frac{-x(k+2) + 8x(k+1) - 8x(k-1) + x(k-2)}{12\Delta t} \tag{1}$$

$$a(k) = \frac{-x(k+2) + 16x(k+1) - 30x(k) + 16x(k-1) - x(k-2)}{12(\Delta t)^2} \tag{2}$$

To identify walking periods, horizontal COM velocity was computed as the Euclidean norm of ML and AP COM velocities. The velocity signal was low-pass filtered using a zero-phase 4th-order Butterworth filter with a cutoff frequency of 4 Hz. A preliminary walking mask was obtained by thresholding the filtered speed at 0.12 m/s. Short segments were then removed by enforcing a minimum walking duration of 0.8 s and a minimum stop duration of 0.5 s. To improve robustness near transitions, the walking mask was expanded by a 0.30 s margin (morphological dilation implemented via convolution). Finally, only walking bouts with duration longer than 3.0 s were retained for entropy analysis.

COM accelerations in the ML and AP directions were obtained from the position signals using the four-point finite-difference scheme described above. Because numerical differentiation amplifies high-frequency measurement noise, the resulting acceleration time series were low-pass filtered using a zero-phase 4th-order Butterworth filter with 20 Hz cutoff frequency to attenuate high-frequency noise while preserving gait-related signal content. A comparable cutoff has been used for preprocessing trunk/walking acceleration signals prior to downstream feature extraction. [11]

For each trial, normalization statistics were computed from samples within the retained walking bouts only. Specifically, the mean and standard deviation were calculated from the acceleration samples within the final walking mask, and each acceleration signal was z-scored using these values.

2.5. Sample Entropy Computation

Sample entropy (SampEn) was used to quantify the temporal irregularity of the center-of-mass (CoM) acceleration signals in the ML and AP directions. For each retained walking bout, the acceleration time series was first normalized using z-scoring to remove inter-trial amplitude differences. SampEn was then computed using an overlapping sliding-window approach, with a window length of 3s and 80% overlap. All valid window-level SampEn values within each trial were collected and exported for downstream condition-level comparisons.

SampEn was defined as the negative logarithm of the conditional probability that two sequences that are similar for m consecutive samples remain similar when the sequence length increases to m+1, excluding self-matches:

$$SampEn(m, r, N) = -\ln\left[\frac{A^{(m+1)}(r)}{B^{(m)}(r)}\right] \quad (3)$$

where $B^{(m)}(r)$ denotes the proportion of matched template pairs of length m within tolerance r, and $A^{(m+1)}(r)$ denotes the corresponding proportion for templates of length m+1. Following common practice in physiological time-series analysis, parameters were set to $m = 2$ and $r = 0.2$ (with expressed as a fraction of the signal standard deviation). SampEn computation followed the standard definition proposed by Richman and Moorman. [10]

2.6. Statistical Analysis

A linear mixed-effects framework was used to test whether SampEn differed across exoskeleton modes while accounting for repeated measurements within subjects (random intercept for Subject) [12]. For each direction (ML and AP), an omnibus fixed-effect test was first performed to evaluate the overall exoskeleton effect. When the omnibus test indicated a potential exoskeleton effect, post-hoc pairwise contrasts between exoskeleton were conducted on the model's estimated marginal means. To control the family-wise error rate introduced by multiple pairwise testing, Holm's step-down procedure was applied to adjust p-values; this approach is less conservative than the classic Bonferroni correction while still maintaining strong control of Type I error.

3. Results

Figure 2 summarizes the model-estimated marginal means (EMMs) for each exoskeleton with ±1 standard deviation. Compared with the no-exoskeleton (NE) condition, the exoskeleton conditions showed a trend toward lower SampEn in both ML and AP directions. Because SampEn quantifies the irregularity of a time series, lower SampEn reflects more regular, less complex acceleration patterns, suggesting more consistent gait-related COM acceleration dynamics under exoskeleton assistance.

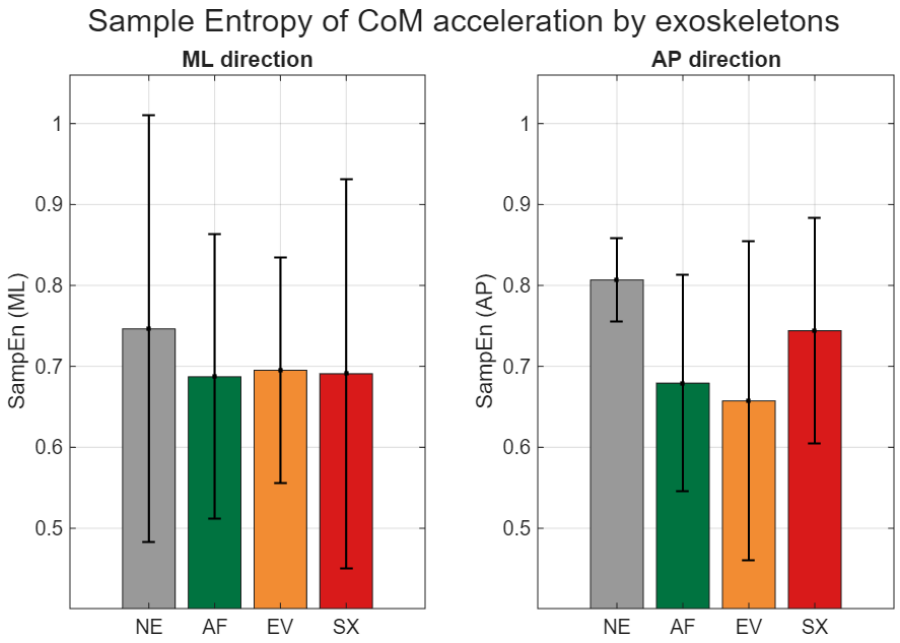


Figure 2: Sample entropy of acceleration of COM in ML and AP direction among No exoskeleton and 3 different exoskeletons after using LME model.

As Table 1 shows below, using this analysis pipeline, no evidence of a Exo-related difference was observed in the ML direction, as the omnibus test was not significant and the post-hoc contrasts did not identify any reliable between-exo separations after Holm adjustment. (significance level $\alpha = 0.05$)

Table 1: Holm-adjusted pairwise comparison of SampEn in ML direction.

Comparison	Diff	t	$p_{uncorrected}$	p_{Holm}	Significance
NE vs AF	0.0591	0.8175	0.4199	1	No
NE vs EV	0.0511	0.7336	0.4687	1	No
NE vs SX	0.0555	0.7963	0.4319	1	No
AF vs EV	-0.0080	-0.1108	0.9125	1	No
AF vs SX	-0.036	-0.0504	0.9601	1	No
EV vs SX	0.0044	0.0627	0.9504	1	No

In the AP direction, no significant overall exoskeleton effect was observed. Post-hoc pairwise comparisons also showed no statistically significant differences between conditions after multiplicity adjustment in Table 2. Nevertheless, a descriptive trend toward lower SampEn under the EV condition relative to NE was apparent, representing the largest separation among the AP comparisons. (significance level $\alpha = 0.05$)

Table 2: Holm-adjusted pairwise comparison of SampEn in AP direction.

Comparison	Diff	t	$p_{uncorrected}$	p_{Holm}	Significance
NE vs AF	0.1275	2.1039	0.0436	0.2180	No
NE vs EV	0.1494	2.5487	0.0160	0.0959	No
NE vs SX	0.0627	1.069	0.2933	0.8789	No
AF vs EV	0.0219	0.3617	0.7201	0.8789	No
AF vs SX	-0.0648	-1.0698	0.2930	0.8789	No
EV vs SX	-0.0868	-1.4797	0.1490	0.5962	No

Although no statistically significant differences were detected across exoskeleton conditions, the AP direction showed the largest descriptive separation in the regularity of COM acceleration, with the clearest contrast observed between the no-exoskeleton condition and the Ekso Bionics Ekso Vest condition..

4. Discussion and Limitations

This work proposes a task-sequence evaluation framework for occupational exoskeletons using a simulated manual material handling protocol and nonlinear time-series analysis of whole-body dynamics. While many exoskeleton studies focus on local biomechanical outcomes during isolated lifting, real workflows often require immediate transitions between handling and locomotion. Such transitions may involve changes in whole-body control strategies that are not fully captured by static or quasi-static metrics. By analyzing COM acceleration during walking, the present framework targets a global indicator of gait control relevant to forward progression and dynamic balance.

The use of sample entropy is motivated by the view that gait stability and adaptability are reflected not only in the magnitude of variability but also in the temporal organization of the signal. Changes in SampEn of COM acceleration may reflect altered regularity of locomotor control associated with exoskeleton use, potentially through changes in posture, segment coordination, or sensorimotor strategy. The proposed analysis is therefore suitable for examining possible condition-dependent shifts in gait dynamics following repeated lift–carry–place cycles.

Although no statistically significant differences were detected across exoskeleton conditions in either the AP or ML direction, the AP results showed the largest descriptive separation between NE and EV. This pattern suggests that exoskeleton-related changes in the temporal regularity of locomotor control may still be present, but the current pilot sample was insufficient to confirm such differences statistically. Therefore, the present findings should be interpreted cautiously and not as evidence that one device is superior based on stability outcome alone. In real occupational settings, exoskeleton adoption also depends on perceived comfort, usability, task compatibility, training burden, and cost, so stability-oriented results should be considered together with broader biomechanical and implementation-related factors.

This preliminary analysis includes 9 participants, which limits statistical power and the generalizability of the findings. Future work should include a larger cohort and consider stratification by sex, body size, prior experience with exoskeleton use, and subjective measures such as comfort and usability.

5. Conclusion

This study evaluated the regularity of COM acceleration during walking across four exoskeleton conditions using sample entropy and linear mixed-effects modeling. No statistically significant differences in SampEn were detected across conditions in either the ML or AP direction. Although the AP direction showed the largest descriptive separation, especially between NE and EV, the current pilot sample was insufficient to confirm these differences statistically. Future work with larger samples, standardized trial lengths, and additional biomechanical and subjective measures is needed to clarify how exoskeleton use may influence functional stability.

Acknowledgement

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